

PROBLEM STATEMENT

Quotient Rule & Composite Chain Rule

Find the derivative with respect to x of the function:

$$y = \frac{\cos^2 x}{\cos(x^2)}$$

STEP 1 প্রদত্ত ত্রিকোণমিতিক ভগ্নাংশ ফাংশনটি উল্লেখ করি (Given Function)

$$y = \frac{\cos^2 x}{\cos(x^2)} = \frac{(\cos x)^2}{\cos(x^2)}$$

STEP 2 ভাগ বিধি (Quotient Rule) অনুসারে অন্তরীকরণ সূত্র স্থাপন করি (Setting up Quotient Rule)

$$\frac{dy}{dx} = \frac{\cos(x^2) \cdot \frac{d}{dx}(\cos^2 x) - \cos^2 x \cdot \frac{d}{dx}[\cos(x^2)]}{[\cos(x^2)]^2}$$

STEP 3 লবের পদ $u = \cos^2 x$ -এর অন্তরজ নির্ণয় করি (Differentiating Numerator via Power & Chain Rule)

$$\frac{d}{dx}(\cos^2 x) = 2 \cos x \cdot \frac{d}{dx}(\cos x) = 2 \cos x (-\sin x) = -2 \sin x \cos x = -\sin 2x$$

STEP 4 হরের পদ $v = \cos(x^2)$ -এর অন্তরজ নির্ণয় করি (Differentiating Denominator via Chain Rule)

$$\frac{d}{dx}[\cos(x^2)] = -\sin(x^2) \cdot \frac{d}{dx}(x^2) = -\sin(x^2) \cdot (2x) = -2x \sin(x^2)$$

STEP 5 মানগুলি সূত্রে স্থাপন করে বীজগাণিতিক সরলীকরণ করি (Substituting into Formula & Simplifying)

$$\frac{dy}{dx} = \frac{\cos(x^2)[-2 \sin x \cos x] - \cos^2 x[-2x \sin(x^2)]}{\cos^2(x^2)} = \frac{2x \cos^2 x \sin(x^2) - \sin 2x \cos(x^2)}{\cos^2(x^2)}$$

FINAL ANSWER

$$\frac{dy}{dx} = \frac{2x \cos^2 x \sin(x^2) - \sin 2x \cos(x^2)}{\cos^2(x^2)}$$

BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- Quotient Rule:** $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{vu' - uv'}{v^2}$.
- Contrasting Composite Rules:**
$$\frac{d}{dx} [(\cos x)^2] = -2 \sin x \cos x = -\sin 2x \quad \text{vs} \quad \frac{d}{dx} [\cos(x^2)] = -2x \sin(x^2)$$
- Double Angle Formula:** $2 \sin x \cos x = \sin 2x$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- Confusion Between Functional Square & Argument Square:** Treating $\cos^2 x$ and $\cos(x^2)$ as identical. They have fundamentally different derivative behaviors.
- Double Negative Sign in Quotient Rule:** $-u \cdot v' = -\cos^2 x \cdot (-2x \sin(x^2)) = +2x \cos^2 x \sin(x^2)$. Dropping a minus sign leads to an erroneous minus between the terms.
- Cancelling Cosines Illegally:** Cannot cancel $\cos(x^2)$ from the denominator with $\cos x$ in the numerator as their arguments are distinct.

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- Denominator Zero Restriction:** $\cos(x^2) \neq 0 \implies x^2 \neq \frac{\pi}{2} + k\pi$ for $k \geq 0, k \in \mathbb{Z}$.
- Singular Points:** Vertical asymptotes occur at $x = \pm \sqrt{(2k+1)\frac{\pi}{2}}$.
- Differentiability:** Smooth and differentiable on all open intervals between the consecutive roots of $\cos(x^2) = 0$.

BOX 4: LOGARITHMIC DIFFERENTIATION ALTERNATIVE

- Taking natural log: $\ln y = 2 \ln(\cos x) - \ln(\cos(x^2))$
- Differentiating implicitly:
$$\frac{1}{y} \frac{dy}{dx} = 2 \left(\frac{-\sin x}{\cos x} \right) - \left(\frac{-\sin(x^2) \cdot 2x}{\cos(x^2)} \right) = -2 \tan x + 2x \tan(x^2)$$
- Multiplying by $y = \frac{\cos^2 x}{\cos(x^2)}$:
$$\frac{dy}{dx} = \frac{\cos^2 x}{\cos(x^2)} [2x \tan(x^2) - 2 \tan x] = \frac{2x \cos^2 x \sin(x^2) - 2 \sin x \cos x \cos(x^2)}{\cos^2(x^2)}$$

BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- Step 1-2 (Quotient Rule Structure Setup):** 1.0 Mark
- Step 3 (Numerator Derivative with Chain Rule):** 1.0 Mark
- Step 4 (Denominator Derivative with Chain Rule):** 1.0 Mark
- Step 5 & Final Answer:** 1.0 Mark
- Total Allocation:** 4.0 Marks